Approximate Computing		
Paderborn University		SoSe 2026
15.04.2026		Deadline: 16.07.2026
Lecturers: **Dr. Muhammad Awais** **Prof. Dr. Marco Platzner**		**Lab Module 2:** **Task**

1 Objective

Get familiar with the process of synthesis and evaluation of approximate circuits. Here, you develop a very simple framework to optimize the area of the given circuit, i.e. a DSP filter, using approximate components.

2 Input Circuit

The input circuit for this given task is an FIR filter as given in Figure 1. FIR filters are one of the most common digital filters used. You have to write the code that implements a 10 stage FIR filter in VHDL (Verilog) subject to the constraints. For more information on FIR filter you can have a look at:

`https://en.wikipedia.org/wiki/Finite_impulse_response`

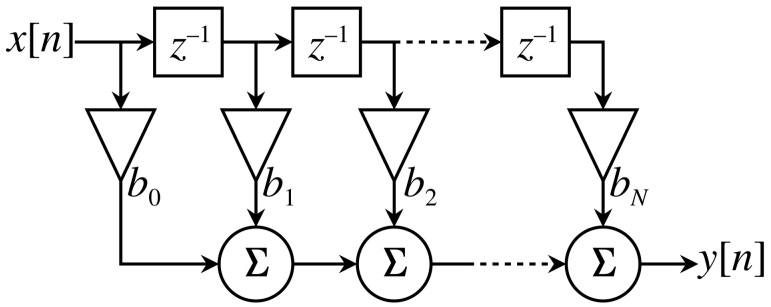


Figure 1: Target FIR circuit

2.1 FIR Specifications

- Inputs: 1 32-bit number (`std_logic_vector`)
- Output: 1 32-bit number (`std_logic_vector`)
- Function: 10-stage FIR

- Constants:
  - $b_0 = b_9 = 3$
  - $b_1 = b_8 = 25$
  - $b_2 = b_7 = 96$
  - $b_3 = b_6 = 384$
  - $b_4 = b_5 = 3073$
- Carry and overflows for additions and multiplications are ignored

2.2 Constraints

- Timing (for synthesis): clock cycle 1.0 ns
- Throughput: 1 result per cycle
- Latency: ≤ 3 cycles

3 Approximate Operations

- Adders: MSB truncated from 1 to 8 bit(s), LSB truncated from 1 to 8 bit(s)
- Multipliers: MSB truncated from 1 to 8 bit(s), LSB truncated from 1 to 8 bit(s)

4 Design Space Exploration Algorithm

The target algorithm for DSE is beam search. Beam search is a heuristic search algorithm that explores a given graph by expanding the most promising node (here area saving) in a limited set. Here, the set size is 3, which means that in every iteration, the top 3 approximate instances are selected for further expansion.

4.1 Optimization Goal

The optimization goal here is the circuit area. When the search stops, all the valid nodes, i.e., nodes that do not violate the Mean Squared Error of 5% on the testing set, should be sorted based on their estimated area, and the top 10

are selected. The actual area of these candidates should be calculated by the synthesis tool, i.e, Synopsis Design Compiler.

4.1.1 Area Estimation

The area estimation can be done to prevent running costly circuit synthesis. You need to precompute the area of each approximate component before running the design space exploration. Every time you want to evaluate the area, it could be estimated by the sum of the areas of its components.

5 Quality Evaluation

Quality evaluation is performed via testing. Your testset must contain $10,000$ random numbers from a uniform distribution in the range of $[0, 255]$ without repetition. The quality of each node that is added to the search tree should be evaluated, and those that have an MSE larger than 5% should be dropped from further consideration.

6 Framework implementation

For components, target circuit, and the testing process, you may use VHDL, Verilog, or even SystemC. The approximate algorithm can be implemented in any high-level language, e.g., Python or C++.

7 Report

You should provide a brief explanation, in PDF format, about the architecture of your framework as well as the results you obtained. The report should contain a Pareto space graph for the obtained top 10 approximate instances.